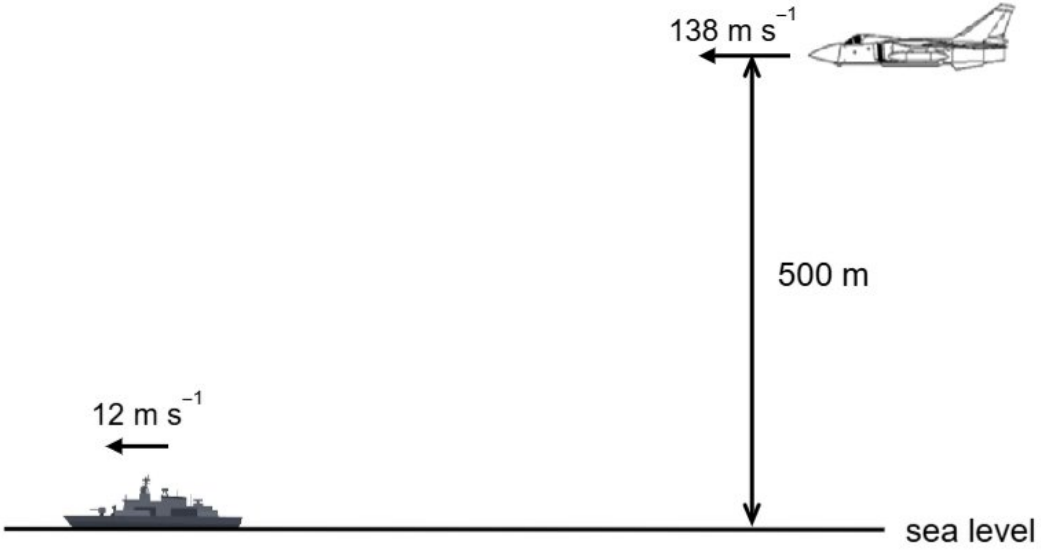
3 A fighter bomber is travelling at 138 m s^{-1} horizontally at a height of 500 m above sea level. It is on a bombing run to its target, a naval ship which is moving at 12 m s^{-1} away from the bomber.



Assume negligible air resistance.

If the pilot wants to score a direct hit, at which position should he release the bomb?

A When the plane is 1010 m away from the naval ship horizontally.

B When the plane is 1270 m away from the naval ship horizontally.

C When the plane is 1390 m away from the naval ship horizontally.

D When the plane is 1510 m away from the naval ship horizontally.